

Classifying Reported Diabetes Status in the CCHS 2022: A Literature Review on Predictor Choice and Model Evaluation

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1. Introduction

The Canadian Community Health Survey (CCHS) is a cross-sectional survey run by Statistics Canada. It collects information on health status, use of health care services and health determinants for people aged 12 and older, and part of the data is released as a Public Use Microdata File (PUMF) [1]. The 2022 PUMF used in this project contains 67,079 respondents and 255 variables, and the outcome is CCC_05, self-reported diabetes status.

This project asks how the choice of information given to a model affects how well it can classify which respondents report having diabetes, and how that effect compares with the effect of changing the model itself. This review looks at published diabetes prediction models and the predictors they use, Canadian population-level work, the evidence behind individual predictors, and methodological issues that come up when machine learning is applied to survey data. It also explains how this literature guided the feature conditions tested in the project.

2. Diabetes Prediction Models and Their Predictors

Many risk models for type 2 diabetes have been published. Noble et al. [2] reviewed published risk models and scores, combining quantitative and qualitative methods to judge how they could be selected and used in practice. Collins et al. [3] reviewed how these models were developed and reported. They found that the most commonly used predictors were age, family history of diabetes, body mass index (BMI), hypertension, waist circumference and sex, and they also found frequent weaknesses in how models were built and described, such as the handling of missing data. Asgari et al. [4] reviewed models for undiagnosed and future type 2 diabetes in the general population using the TRIPOD and PROBAST frameworks, and again age and BMI were among the most frequently reported predictors.

Machine learning has also been applied to this problem. De Silva et al. [5] carried out a systematic review and meta-analysis of machine learning models for type 2 diabetes prediction in community settings. The pooled C-index was 0.825 for neural networks and 0.815 for logistic regression, which means the simpler model performed almost as well. The authors also noted that the lack of external validation was a barrier to using these models in practice. This finding is directly relevant to the present project, which compares logistic regression with tree-based models on the same features.

3. Canadian Population-Level Work

The most important Canadian comparison is the Diabetes Population Risk Tool (DPoRT) developed by Rosella et al. [6]. DPoRT was built from Canadian health survey responses

linked to a population-based diabetes registry, and it predicts the risk of developing diabetes over nine years. Its sex-specific models use predictors such as BMI, age, ethnicity, hypertension, immigrant status, smoking, education and heart disease. DPoRT is useful for this project because it shows that routinely collected survey variables can support diabetes prediction in the Canadian population.

Later Canadian work has focused on social differences in diabetes risk. Lu, Kornas and Rosella [7] used a validated population risk algorithm to model diabetes risk in Canadian population groups defined by socioeconomic and lifestyle characteristics, and found clear inequities between groups. Earlier, Dinca-Panaitescu et al. [8] used CCHS data and found that the prevalence of type 2 diabetes in the lowest income group was 4.14 times higher than in the highest income group. These studies support the use of income and education as predictors in a Canadian survey setting, although an association with diabetes does not guarantee that a variable will improve a classifier once stronger predictors are already included.

4. Evidence on Individual Predictors

Age and BMI. Age and BMI have the strongest support in the prediction literature [3], [4], [6]. In the CCHS, BMI is recorded with one classification for adults and a different one for respondents aged 12 to 17, so the project combined the two into a single harmonised BMI class.

Hypertension and cholesterol. Hypertension is one of the most common predictors in published models [3] and is part of DPoRT [6]. Reported high cholesterol is a related cardiometabolic condition that is recorded in the CCHS. Both were tested as additions to the compact core set in this project.

Smoking and physical activity. A meta-analysis of 88 prospective studies by Pan et al. [9] found that current smokers had a 37% higher risk of type 2 diabetes than people who had never smoked. Aune et al. [10] found a dose-response relationship in which more physical activity was linked to lower risk of type 2 diabetes. In the CCHS, the relevant smoking and physical activity measures mainly apply to adults, so they were planned as separate adult-only conditions.

Alcohol and vaping. Baliunas et al. [11] found a U-shaped relationship between alcohol intake and type 2 diabetes, which means a simple drinker-type category is a weak way to represent alcohol use. For e-cigarettes, Adebisi et al. [12] found no evidence that exclusive e-cigarette use was associated with diabetes, while people who used both cigarettes and e-cigarettes showed a modest association. For these reasons, alcohol type and ever having tried vaping were not included in the compact core.

Psychosocial factors. Yu et al. [13] found that people with depression had a higher risk of developing diabetes, with a pooled relative risk of 1.41. In a cross-sectional survey, however, low mood or low life satisfaction may also be a result of living with diabetes, so such variables were not treated as core predictors.

5. Methodological Issues

5.1 A cross-sectional outcome

Most of the prediction literature is about future type 2 diabetes [3], [4], [6]. The CCHS records a person's characteristics and their diabetes status at the same time, and CCC_05 covers reported diabetes in general rather than type 2 diabetes only. This project therefore classifies reported diabetes status and does not predict future risk. It also cannot show that any variable causes diabetes. For example, a person's BMI may have changed after diagnosis, so in this setting variables are described as features associated with reported diabetes, not as risk factors.

5.2 Leakage and the diagnosis pathway

Kaufman et al. [14] describe leakage as information in the training data that would not be legitimately available for the prediction task. Variables about diabetes treatment or management were excluded because they follow from the outcome. Having a regular health care provider (RHC_05) was also held out, because diabetes has to be diagnosed by a health professional, so contact with the health system can affect whether diabetes is reported at all. A model using that variable might partly learn who has been diagnosed rather than who has diabetes.

5.3 Survey weights

The CCHS uses a complex sampling design. Kim, Rogers and Wang [15] found that including or leaving out survey weights can change both the variables selected and the measured performance of machine learning models for chronic disease. In this project the sampling weight WTS_M was kept out of the features because it describes the survey design rather than the respondent. The weights were not applied in training or evaluation, so the results describe the survey sample and are not population estimates. This is a limitation of the project.

5.4 Class imbalance, calibration and validation

About 9% of the modelling population reports diabetes, so accuracy alone is misleading. Saito and Rehmsmeier [16] show that precision-recall analysis is more informative than ROC analysis for imbalanced data, so PR-AUC is reported together with ROC-AUC, precision, recall and F1. Van Calster et al. [17] stress that calibration is often ignored in prediction studies, and the reviews in Section 2 point to weak validation as a common problem [4], [5]. In response, this project chose each classification threshold on a separate validation set, checked the stability of the model comparison with 5-fold cross-validation on the training data, and repeated the final comparison on a fresh holdout split. The models were built with scikit-learn [18].

6. How the Literature Shaped the Feature Conditions

The literature was used to test and revise the feature sets rather than to justify a fixed list chosen in advance. Experiment C used a compact core of seven broadly applicable variables: age, sex, education, harmonised BMI class, household income, immigrant status and

province. Condition D1 added reported high blood pressure (CCC_80), and D2 added reported high cholesterol (CCC_90). Smoking and physical activity were planned as adult-only conditions but were not completed, so they remain future work. Table 1 summarises how each predictor family was treated.

Table 1. Predictor families, supporting literature and their use in this project

Predictor family	Evidence in the literature	CCHS 2022 variable	Use in this project
Age	Among the most common predictors [3], [4]	DHHGAGE	Core (Experiment C onward)
BMI	Among the most common predictors [3], [4]	BMI class built from HWTGDGISHW and HWTGDGWHO	Core (Experiment C onward)
Sex	Common in published models [3]	DHH_SEX	Core
Education and income	Canadian population studies [6], [7], [8]	EDDVH3, INCDGHH	Core
Immigrant status	Included in DPoRT [6]	SDCDGIMM	Core, treated as context
Province	Regional context	GEOGPRV	Core, treated as context
Hypertension	Common predictor [3], part of DPoRT [6]	CCC_80	Added in D1
High cholesterol	Related cardiometabolic condition	CCC_90	Added in D2
Smoking	Higher risk in current smokers [9]	Adult smoking measures	Planned adult-only condition, not completed
Physical activity	Dose-response relationship [10]	Adult activity measures	Planned adult-only condition, not completed
Alcohol	U-shaped relationship [11]	ALCDVTTM	Not in the compact core
Vaping	No clear link for exclusive use [12]	ECV_05	Not in the compact core
Regular health care provider	Linked to diagnosis, not to disease	RHC_05	Held out

7. Research Gap and Contribution of This Project

There is a large body of work on diabetes risk prediction, including Canadian population tools and machine learning studies [2], [5], [6]. Less work has looked at how much the choice and representation of predictors matters compared with the choice of algorithm when classifying reported diabetes in the CCHS 2022 PUMF, while handling the survey's special response codes and applicability rules carefully. This is the gap the project addresses.

In the project, ROC-AUC ranged from 0.774 to 0.827 across the completed feature conditions, and F1 changed by 0.053 between conditions C and D2. On the fixed D2 feature set, the F1 scores of logistic regression, random forest and gradient boosting differed by only 0.008. Age, reported hypertension, reported high cholesterol and BMI class were the most important variables for all three models. These results agree with the predictors highlighted in the literature [3], [4], [6], and the small difference between models is consistent with the review by De Silva et al. [5].

8. Limitations of This Review

This was a targeted review rather than a formal systematic review, and it focused on English language sources. Most of the prediction studies reviewed were designed for future type 2 diabetes, which is a different task from classifying reported diabetes in a cross-sectional survey, so their findings were used as guidance rather than as direct benchmarks.

9. Conclusion

The literature strongly supports age, BMI and hypertension as predictors of diabetes, gives reasonable support for sex, education and income in the Canadian setting, and gives weaker or more conditional support for alcohol, vaping and psychosocial measures. It also warns about leakage, class imbalance, survey design and weak validation. This project used that evidence to build and test feature conditions step by step, and its results suggest that the choice of predictors had a larger effect on classification than the choice of model.

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